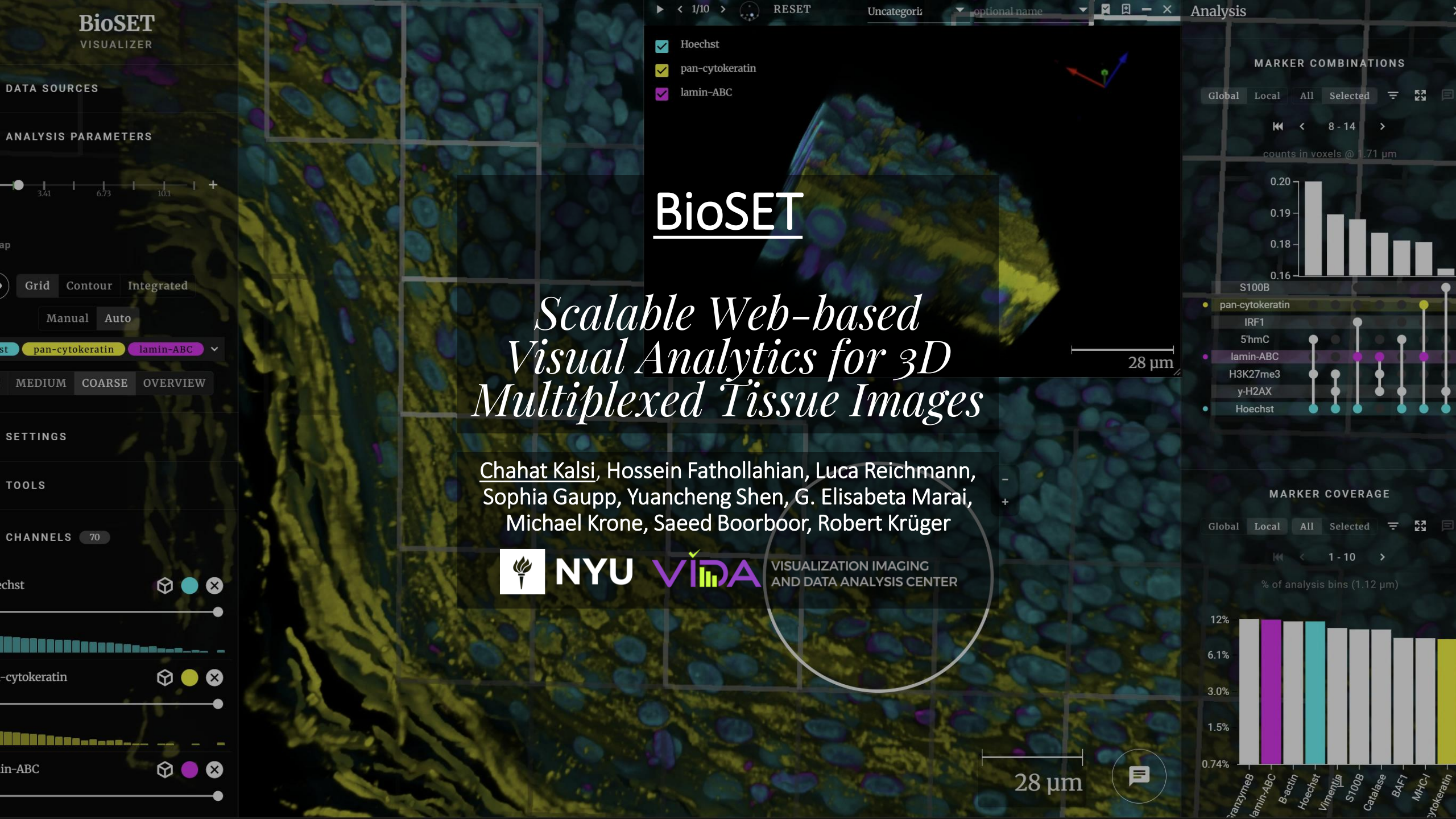




BioSET
VISUALIZER

DATA SOURCES

ANALYSIS PARAMETERS

3.41 6.73 10.1 +

ap

Grid Contour Integrated

Manual Auto

st pan-cytokeratin lamin-ABC

MEDIUM COARSE OVERVIEW

SETTINGS

TOOLS

CHANNELS 70

chost

-cytokeratin

in-ABC

BioSET

*Scalable Web-based
Visual Analytics for 3D
Multiplexed Tissue Images*

Chahat Kalsi, Hossein Fathollahian, Luca Reichmann,
Sophia Gaupp, Yuancheng Shen, G. Elisabeta Marai,
Michael Krone, Saeed Boorboor, Robert Krüger



NYU



VISUALIZATION IMAGING
AND DATA ANALYSIS CENTER

MARKER COMBINATIONS

Global Local All Selected

8 - 14

counts in voxels @ 1.71 μm

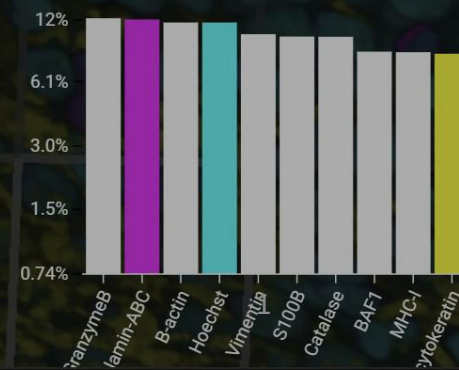


MARKER COVERAGE

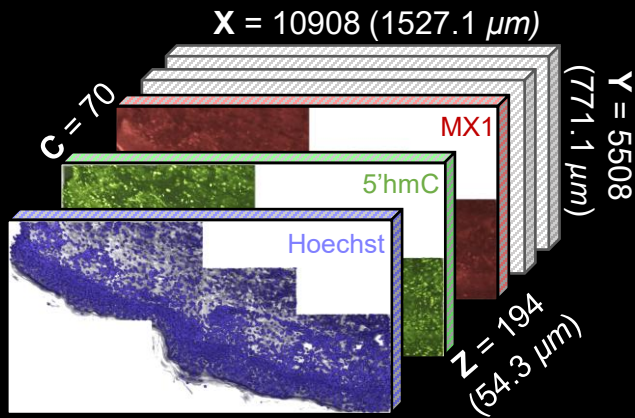
Global Local All Selected

1 - 10

% of analysis bins (1.12 μm)

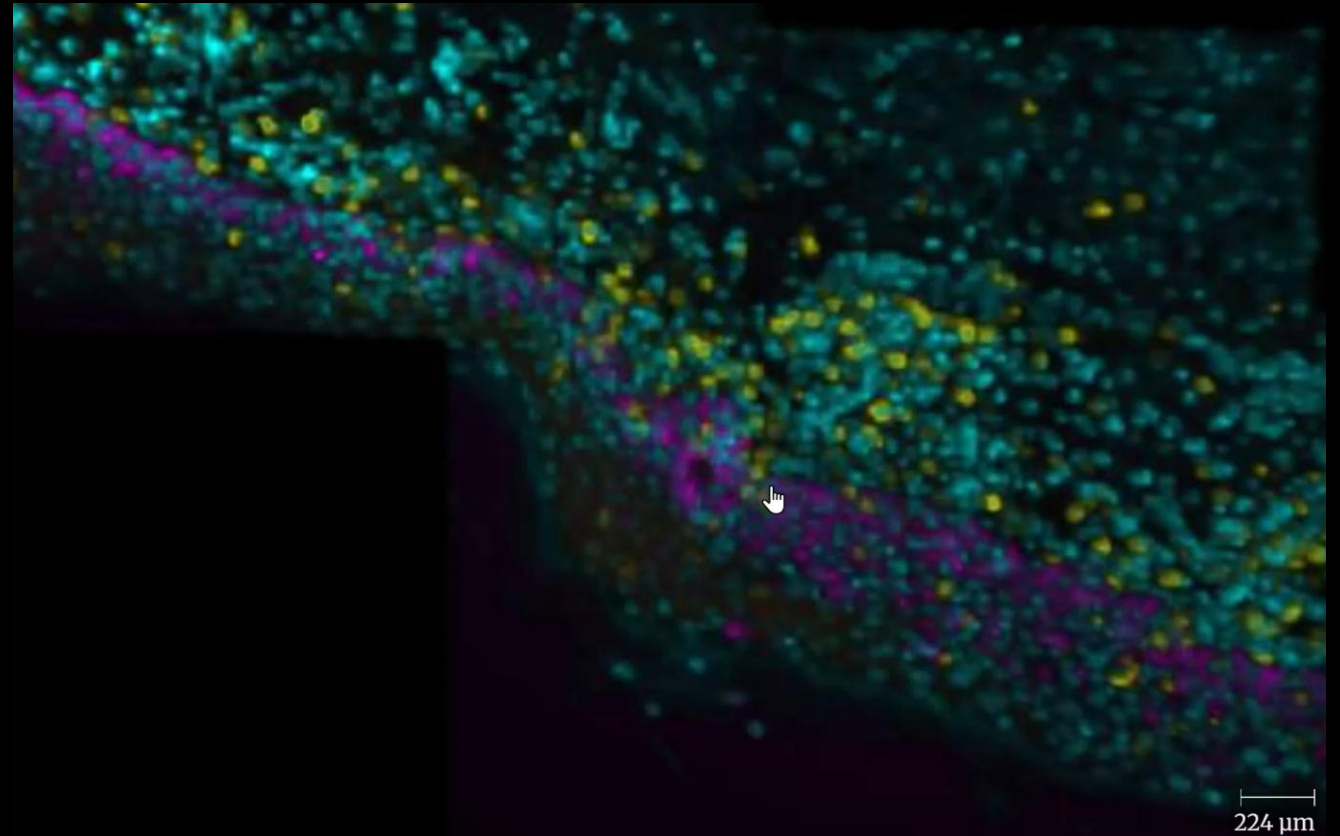


3D Cyclic Immunofluorescence (CyCIF)



Highly multiplexed 3D profiling of cell states and immune niches in human tumors, Yapp et. al, *Nature Methods* (2025)

Melanoma-in-situ: 1.7 TB, 70 Channels



3-Dimensional
30-60 μm thick, ~3 layers of intact cells

Highly Multiplexed
50+ channels

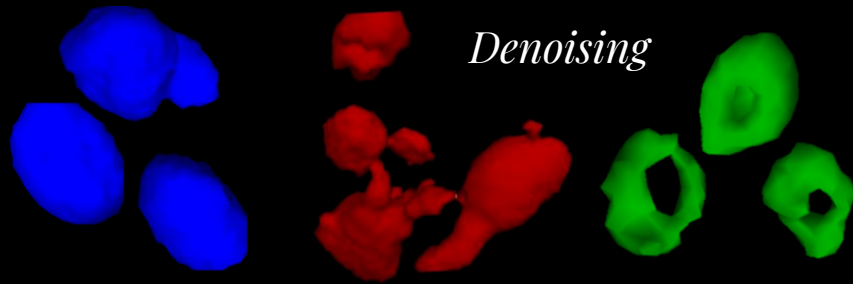
High Resolution
1-3 Terabyte Sized

Challenges to Manual Discovery

Visual Analytics

Guided Human-in-the-loop Discovery

Spatial Biomarker Co-localizations

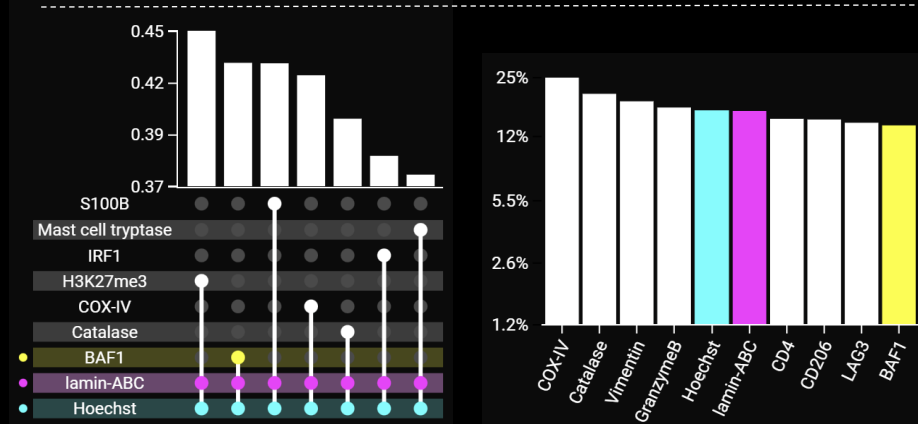


How much are biomarkers
(co-)expressed?

Where in the tissue are they
(co-)expressed?

Plots Summarizing Expressions and Co-expressions

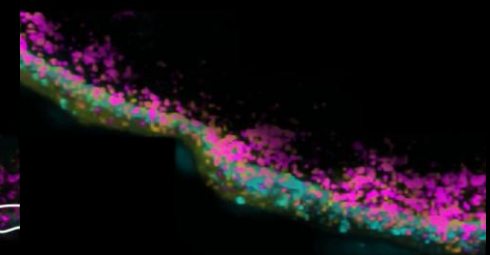
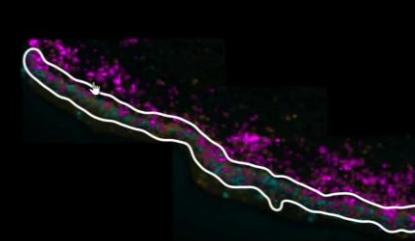
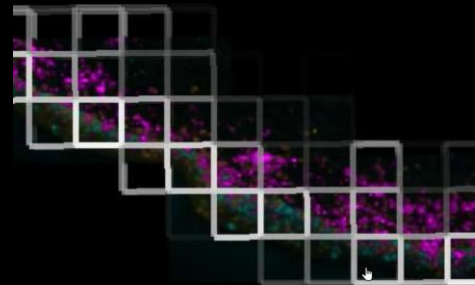
Embedded Spatial Guidance Mechanisms



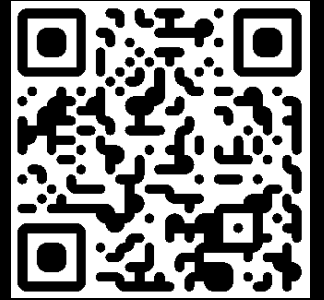
Grid Cells

Outlines

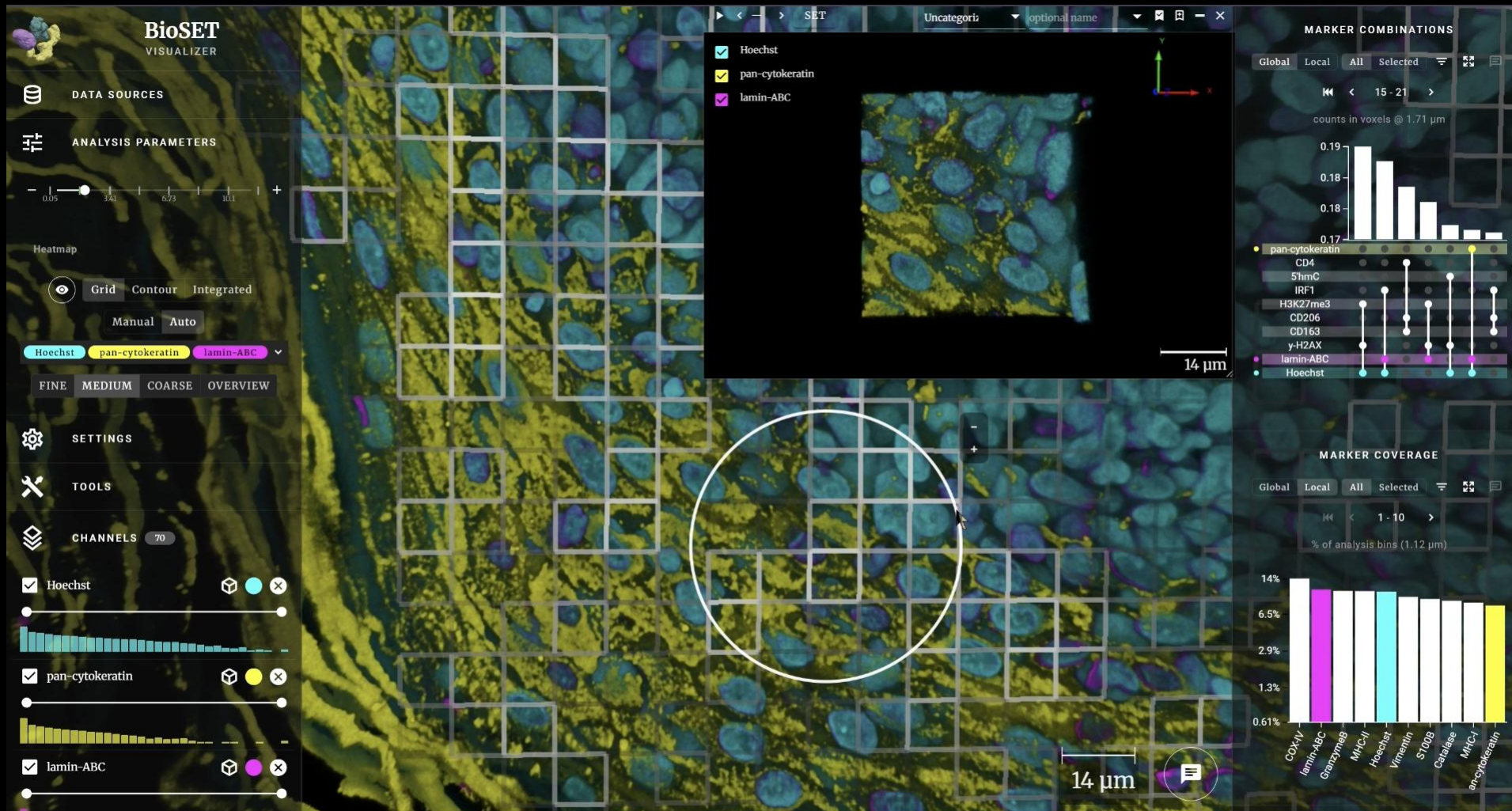
Brightness



BioSET Pipeline and Web Tool



BioSET GitHub



- For scalable processing and web browser visualization of large **terabyte sized 3D tissue images**.
- **Poster Session 2 (Tuesday): Po48** – “Scalable 3D Visualization and Spatial Exploration of Multiplexed Cancer Tissue Images for the Web”
- **Let’s collaborate!** Have a dataset or use case in mind? We’d love to explore how BioSET can support your work.